Timo **Menzel**

Graphics Engineering • Digital Human Systems

I am a PhD candidate and research assistant at TU Dortmund University, specializing in 3D avatar scanning and personalization using consumer hardware. I build pipelines that turn smartphone captures into realistic full-body avatars and personalized facial animations for VR and real-time applications. I aim to bring my expertise in graphics engineering and real-time systems to industry, contributing to computer graphics, digital humans, or interactive 3D projects.

Contact

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Skills

Graphics & Vision

Geometry Processing · Photogrammetry · Non-rigid Registration · Kinematics · Blendshapes · Mesh Reconstruction

Programming

C++ · Swift · Python · C#

Frameworks

ARKit · OpenCV · Eigen · RealityKit · MediaPipe · OpenGL · iOS · macOS · Linux · GoogleTest · XCTest

Tools

Git · Unity · SQL · CMake · GCC

Systems

3D Pipelines · Client-Server Systems

Languages

German	(Native)
English	(Fluent)
French	(Rudimentary)

Experience

Graphics & Geometry Group — *Bielefeld University → TU Dortmund University*

Research Assistant / PhD Candidate — *May 2020 - Present*

- Work continuously with the same research group, which relocated from Bielefeld University to TU Dortmund University in 2020.
- Conduct research in 3D avatar scanning and personalization from smartphone data
  - Develop automated pipelines for avatar reconstruction projects.
  - Collaborate with interdisciplinary teams to evaluate new methods and system usability.
  - Publish findings from current work in peer-reviewed venues.

Selected Projects

Automated Avatar Reconstruction Pipeline — *2023 - 2025*

- Designed and implemented a fully automated end-to-end pipeline for reconstructing VR-ready full-body avatars from smartphone captures.
- Built mobile capture app and scalable server-side reconstruction pipeline, enabling non-expert users to acquire high-quality scans in uncontrolled environments.
- Integrated guided capture, segmentation, landmark detection, photogrammetry, and template fitting into a workflow producing high-quality avatars in 20 minutes.
- Evaluations showed avatars comparable to expert-operated scans, demonstrating feasibility of accessible, large-scale avatar creation.
- **Publication: "Avatars for the Masses: Smartphone-Based Reconstruction of Humans for Virtual Reality"** – `https://doi.org/10.3389/frvir.2025.1583474`
- **Project Page:** `https://avatars.cs.tu-dortmund.de`

Motion Artifact Compensation — *2024 - 2025*

- Extended the "Avatars for the Masses" pipeline to reduce motion-induced artifacts during sequential smartphone capture, improving reconstruction robustness.
- Developed sub-scan processing strategy dividing image sequences into temporal segments, integrating silhouette constraints and inverse rendering.
- Improved accuracy in articulated regions (e.g., arms) while preserving a simple, user-friendly capture process.
- **Publication: "Compensating Motion-Induced Errors in Smartphone-Based VR Avatar Reconstruction"** – `https://doi.org/10.1145/3756884.3765995`

Automated Blendshape Personalization for Avatars — *2021 - 2022*

- Developed automated pipeline for personalized facial blendshapes from smartphone-captured expressions using ARKit input.
- Generated personalized blendshape masks through position-based optimization and implanted them into realistic full-body characters using seamless partial deformation transfer.
- The resulting personalized blendshapes are significantly more accurate in reconstructing facial expressions compared to prior methods.
- **Publication: "Automated Blendshape Personalization for Faithful Face Animations Using Commodity Smartphones"** – `https://doi.org/10.1145/3562939.3565622`

Education

TU Dortmund University
Dortmund, Germany

PhD in Computer Science *2020 - Expected Summer 2026*

- Research focus on 3D avatar reconstruction and personalization from smartphone data.
- Published multiple first- and shared first-author papers in ACM VRST and Frontiers in VR.
- Developed automated pipelines for avatar reconstruction and blendshape personalization for non-expert users.

Bielefeld University
Bielefeld, Germany

M.Sc. in Intelligent Systems *2018-2020*

- Interdisciplinary program covering AI, intelligent systems, and human-technology interaction.
- Focus on developing and deploying adaptive, interactive systems using modern software engineering and scientific methods.

Bielefeld University
Bielefeld, Germany

B.Sc. in Informatics in the Natural Sciences *2015-2018*

- Interdisciplinary computer science curriculum focused on applying computational methods to natural science problems (physics, chemistry, biology).
- Emphasis on scientific computing, data analysis, and model-driven software development for research and industry contexts.

Publications

Friedemann Runte, Timo Menzel, Ulrich Schwanecke, and Mario Botsch. *Compensating Motion-Induced Errors in Smartphone-Based VR Avatar Reconstruction*. In: *Proceedings of the 2025 31st ACM Symposium on Virtual Reality Software and Technology*. VRST '25. 2025.
DOI: `10.1145/3756884.3765995`.

Timo Menzel, Erik Wolf, Stephan Wenninger, Niklas Spinczyk, Lena Holderrieth, Carolin Wienrich, Ulrich Schwanecke, Marc Erich Latoschik, and Mario Botsch. *Avatars for the Masses: Smartphone-Based Reconstruction of Humans for Virtual Reality*. In: *Frontiers in Virtual Reality* 6 (2025).
DOI: `10.3389/frvir.2025.1583474`.

Peter Kullmann, Theresa Schell, Timo Menzel, Mario Botsch, and Marc Erich Latoschik. *Coverage of Facial Expressions and Its Effects on Avatar Embodiment, Self-Identification, and Uncanniness*. In: *IEEE Transactions on Visualization and Computer Graphics* 31 (2025).
DOI: `10.1109/TVCG.2025.3549887`.

Peter Kullmann, Timo Menzel, Mario Botsch, and Marc Erich Latoschik. *An Evaluation of Other-Avatar Facial Animation Methods for Social VR*. in: *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. CHI EA '23. 2023.
DOI: `10.1145/3544549.3585617`.

Timo Menzel, Mario Botsch, and Marc Erich Latoschik. *Automated Blendshape Personalization for Faithful Face Animations Using Commodity Smartphones*. In: *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology*. VRST '22. 2022.
DOI: `10.1145/3562939.3565622`.